Traffic Engineering

METI

Lab 2 Report [EN]

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Group 5

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Contents

1 Introduction

The objective of this laboratory assignment was to implement and analyze IP routing mechanisms using Cisco routers within the GNS3 simulation environment.

The experiments performed throughout the laboratory focused on dynamic routing protocols, routing behavior, path selection mechanisms, and routing convergence.

The laboratory activities included:

- IPv4 subnetting using VLSM;
- router interface configuration;
- OSPF routing configuration;
- routing table analysis;
- OSPF metric manipulation;
- connectivity verification;
- link failure simulation and convergence analysis.

The network topology consisted of five routers (R1 to R5) and two end devices (PC1 and PC2), interconnected through multiple redundant paths.

2 Part A — OSPF Network Configuration

2.1 Network Topology

The network topology implemented in GNS3 is presented below.

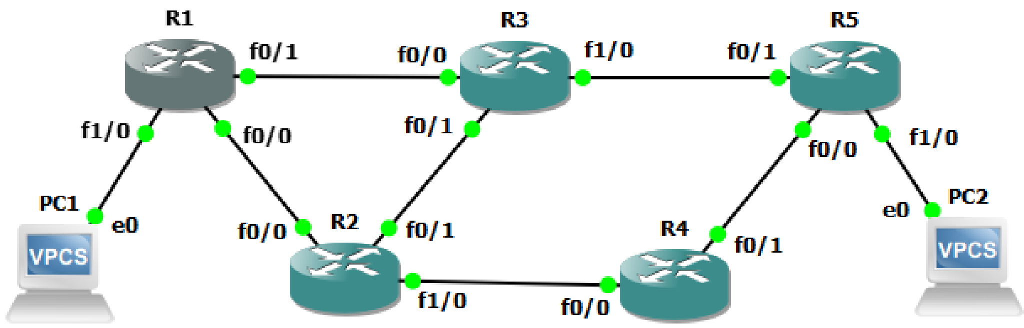


Figure 1: General network topology

The topology contains five routers interconnected through multiple redundant paths, allowing OSPF to dynamically calculate alternative routes in case of failures.
Additionally:

- PC1 is connected to R1;
- PC2 is connected to R5.

2.1.1 Broadcast Domains

Each physical link corresponds to an independent subnet/broadcast domain.
The network contains:

- 6 router-to-router subnets;
- 2 LAN subnets (PC1-R1 and PC2-R5).

Therefore, the topology contains a total of:

8 broadcast domains

2.2 IPv4 Addressing Plan

The network address block assigned was:

192.168.5.0/24

Using VLSM, the address space was segmented into multiple /30 subnets.

Link	**Subnet**	**IPs Used**
R1-R2	192.168.5.4/30	.5 / .6
R1-R3	192.168.5.8/30	.9 / .10
R2-R3	192.168.5.12/30	.13 / .14
R2-R4	192.168.5.16/30	.17 / .18
R4-R5	192.168.5.20/30	.21 / .22
R3-R5	192.168.5.24/30	.25 / .26
PC1-R1	192.168.5.0/30	.1 / .2
PC2-R5	192.168.5.28/30	.29 / .30

Table 1: IPv4 addressing plan

2.3 Router Configuration

Each router interface was configured manually.
Example configuration used on R1:

```
interface FastEthernet1/0
ip address 192.168.5.2 255.255.255.252
no shutdown

interface FastEthernet0/0
ip address 192.168.5.5 255.255.255.252
no shutdown

interface FastEthernet0/1
ip address 192.168.5.9 255.255.255.252
no shutdown
```

The command below was used to verify interface status:

```
show ip interface brief
```

All interfaces reached the `up/up` operational state.
The same configuration methodology was applied to all routers in the topology.

2.4 OSPF Configuration

OSPF was configured using process ID 1.

The following commands were applied:

```
router ospf 1
network 192.168.5.0 0.0.0.255 area 0
```

After configuration, OSPF neighbor relationships were successfully established.

The command used to verify adjacency status was:

```
show ip ospf neighbor
```

Routers reached the FULL adjacency state, confirming successful synchronization of OSPF databases.

All routers were configured within OSPF Area 0.

2.5 Routing Table Analysis

The routing table was inspected using:

```
show ip route
```

Routes learned dynamically through OSPF were identified with the letter:

$$O$$

Initially, multiple equal-cost paths existed for some destinations, demonstrating Equal Cost Multi-Path (ECMP) behavior.

Example:

```
O 192.168.5.12 [110/20] via 192.168.5.10
                        via 192.168.5.6
```

This indicates that two paths with identical OSPF cost were available.

This demonstrated the capability of OSPF to support load balancing across equal-cost paths.

2.6 OSPF Cost Manipulation

To analyze OSPF path selection behavior, the cost of interface FastEthernet0/1 on R1 was increased.

Configuration used:

```
interface FastEthernet0/1
ip ospf cost 100
```

After modifying the cost:

- OSPF recalculated the routing table;
- alternative lower-cost paths were selected;
- traffic was redirected automatically.

Traceroute confirmed the path modification.
Before the cost modification, traffic preferred the direct path:
R1 -¿ R3 -¿ R5
After increasing the OSPF cost on the R1-R3 interface, traffic was redirected through the alternative lower-cost path:
R1 -¿ R2 -¿ R3 -¿ R5
This demonstrated OSPF metric-based path selection.
This experiment demonstrated how OSPF metrics directly influence path selection decisions.

2.7 Experimental Results

This section presents the experimental results obtained during the implementation and testing of the network.

2.7.1 Interface Verification

After configuring the interfaces, the following command was used to verify interface status:

```
show ip interface brief
```

Example output from R1:

```
Interface             IP-Address      Status      Protocol

FastEthernet0/0       192.168.5.5     up          up
FastEthernet0/1       192.168.5.9     up          up
FastEthernet1/0       192.168.5.2     up          up
```

The output confirmed that all interfaces were correctly configured and operational.

2.7.2 OSPF Neighbor Adjacency

The following command was used to verify OSPF neighbor relationships:

```
show ip ospf neighbor
```

Observed output:

```
Neighbor ID      Pri   State       Dead Time    Address         Interface

192.168.5.25       1   FULL/BDR    00:00:32     192.168.5.10    FastEthernet0/1
192.168.5.17       1   FULL/DR     00:00:33     192.168.5.6     FastEthernet0/0
```

The FULL state confirmed that OSPF adjacencies were successfully established and routing databases were synchronized.

2.7.3 Routing Table Analysis

The routing table was analyzed using:

```
show ip route
```

Example output:

```
O 192.168.5.12 [110/20] via 192.168.5.10
                        via 192.168.5.6

O 192.168.5.24 [110/11] via 192.168.5.10

O 192.168.5.16 [110/11] via 192.168.5.6
```

Routes marked with the letter `O` correspond to routes learned through OSPF.

Initially, some routes had multiple equal-cost paths, demonstrating Equal Cost Multi-Path (ECMP) behavior.

2.7.4 OSPF Cost Manipulation

To influence path selection, the OSPF cost of interface FastEthernet0/1 on R1 was increased.

Configuration used:

```
interface FastEthernet0/1
ip ospf cost 100
```

After changing the cost, the routing table changed accordingly.

Updated routing example:

```
O 192.168.5.12 [110/20] via 192.168.5.6

O 192.168.5.24 [110/21] via 192.168.5.6

O 192.168.5.20 [110/21] via 192.168.5.6
```

This confirmed that OSPF selected the path with the lowest accumulated metric.

2.7.5 Connectivity Verification

Connectivity between end devices was tested using ICMP echo requests.

Command executed:

```
ping 192.168.5.30
```

Observed output:

```
!!!!!
Success rate is 100 percent (5/5)
```

This confirmed successful communication between PC1 and PC2.

Traceroute was also performed:

```
traceroute 192.168.5.30
```

Observed route before OSPF cost manipulation:

```
1 192.168.5.6
2 192.168.5.14
3 192.168.5.26
4 192.168.5.30
```

After increasing the OSPF cost, the selected route changed:

```
1 192.168.5.10
2 192.168.5.26
3 192.168.5.30
```

This demonstrated successful dynamic route recalculation.

2.7.6 Link Failure and OSPF Convergence

To evaluate convergence behavior, one of the active interfaces was disabled.
Command executed:

```
interface FastEthernet0/0
shutdown
```

OSPF immediately detected the failure:

```
%OSPF-5-ADJCHG:
Process 1, Nbr 192.168.5.17 on FastEthernet0/0
from FULL to DOWN
```

The routing table was automatically recalculated and traffic was redirected through alternative paths.
Traceroute after failure:

```
1 192.168.5.10
2 192.168.5.26
3 192.168.5.30
```

The interface was restored using:

```
no shutdown
```

OSPF adjacency was re-established successfully:

```
%OSPF-5-ADJCHG:
Process 1, Nbr 192.168.5.17
from LOADING to FULL
```

This demonstrated:

- automatic fault detection;
- dynamic rerouting;
- successful OSPF convergence.

3 Part B — MPLS Configuration and Analysis

3.1 Introduction

The objective of Part B was to configure and analyze an MPLS-based network using Cisco routers in GNS3.

The experiments focused on:

- enabling MPLS forwarding;
- configuring LDP (Label Distribution Protocol);
- analyzing MPLS forwarding tables;
- observing MPLS labels using Wireshark;
- demonstrating Penultimate Hop Popping (PHP);
- testing MPLS recovery after link failures.

The MPLS network was implemented on top of the OSPF infrastructure configured in Part A.

3.2 MPLS Configuration

Cisco Express Forwarding (CEF) was enabled on all routers to support MPLS operation.

The following commands were applied:

```
ip cef
mpls ip
```

MPLS was enabled on the router interfaces participating in the MPLS core.

Example configuration on R3:

```
interface FastEthernet0/0
mpls ip

interface FastEthernet1/0
mpls ip

interface FastEthernet1/1
mpls ip
```

OSPF was used as the underlying routing protocol to distribute reachability information between routers.

3.3 LDP Neighbor Verification

LDP neighbors were verified using the following command:

```
show mpls ldp neighbor
```

Example output obtained on R3:

```
Peer LDP Ident: 192.168.5.9:0
State: Oper

Peer LDP Ident: 192.168.5.29:0
State: Oper

Peer LDP Ident: 192.168.5.17:0
State: Oper
```

The `Oper` state confirmed that LDP sessions were successfully established and labels were being exchanged between MPLS routers.

3.4 MPLS Interface Verification

MPLS-enabled interfaces were verified using:

```
show mpls interfaces
```

Example output:

```
Interface           IP          Operational
FastEthernet0/0     Yes (ldp)   Yes
FastEthernet1/0     Yes (ldp)   Yes
FastEthernet1/1     Yes (ldp)   Yes
```

This confirmed that MPLS forwarding and LDP were operational on all core links.

3.5 MPLS Forwarding Table Analysis

The MPLS forwarding table was inspected using:

```
show mpls forwarding-table
```

Example output obtained on R3:

```
Local   Outgoing   Prefix
Label   Label

16      Pop Label  192.168.5.4/30
17      Pop Label  192.168.5.0/30
18      Pop Label  192.168.5.16/30
19      Pop Label  192.168.5.20/30
20      Pop Label  192.168.5.28/30
```

The forwarding table demonstrated that MPLS labels were correctly assigned and distributed across the network.

The presence of the `Pop Label` operation confirmed the implementation of Penultimate Hop Popping (PHP).

3.6 Wireshark Analysis and PHP Demonstration

Wireshark captures were performed to analyze MPLS label switching behavior.

Traffic was generated using:

```
ping 192.168.5.30
```

and:

```
trace 192.168.5.30
```

3.6.1 MPLS Labels in the Core Network

Wireshark captures performed on core MPLS links showed packets containing MPLS headers.

The captured packets contained entries similar to:

```
MultiProtocol Label Switching Header
Label: 20
```

This confirmed that MPLS labels were being attached and switched correctly inside the MPLS core.

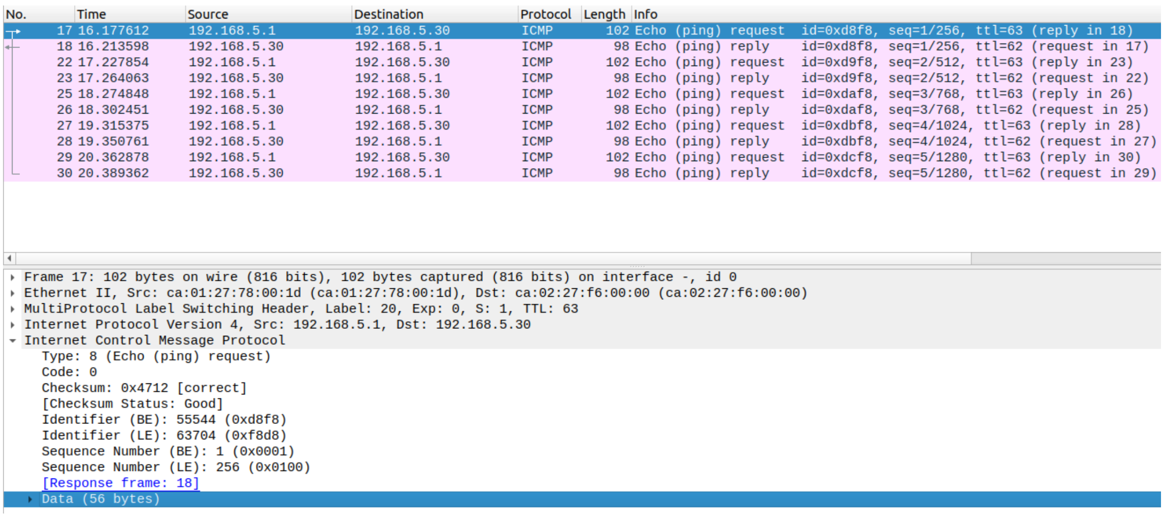


Figure 2: MPLS labeled packet captured in the MPLS core

3.6.2 Penultimate Hop Popping (PHP)

Additional captures were performed on the final hop between R5 and PC2.

The captured packets no longer contained MPLS headers and only displayed IP and ICMP information.

This demonstrated that the MPLS label was removed by the penultimate router before reaching the egress router.

Therefore, Penultimate Hop Popping (PHP) was successfully verified.

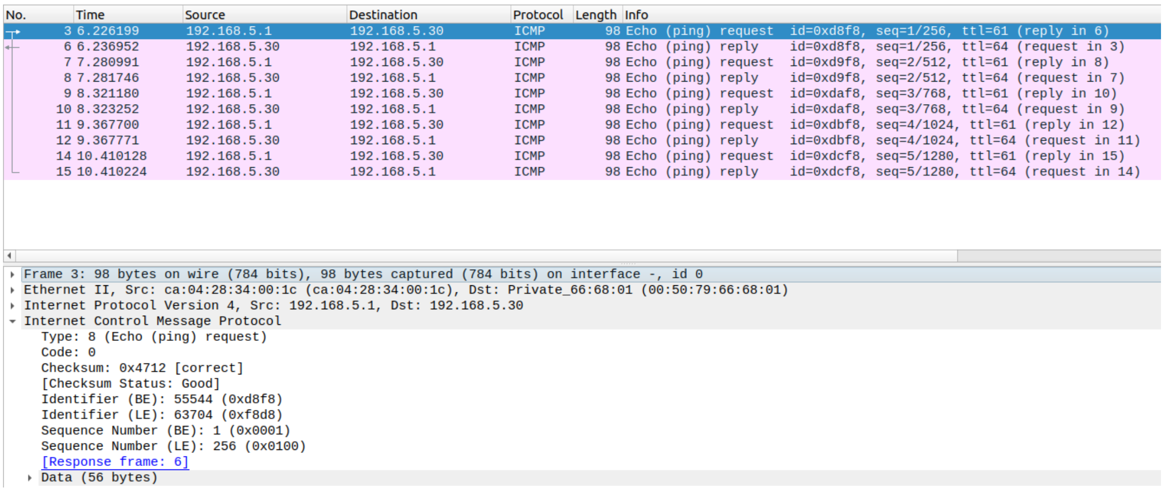


Figure 3: Packet capture after PHP operation (no MPLS label present)

3.7 Connectivity Verification

Connectivity between PC1 and PC2 was verified using ICMP echo requests.

The following command was executed:

```
ping 192.168.5.30
```

Observed output:

```
!!!!!
Success rate is 100 percent (5/5)
```

This confirmed successful MPLS forwarding and end-to-end connectivity.

Traceroute was also performed:

```
trace 192.168.5.30
```

Observed route before link failure:

```
1 192.168.5.2
2 192.168.5.10
3 192.168.5.26
4 192.168.5.30
```

This path corresponded to:

$$R1 \rightarrow R3 \rightarrow R5$$

3.8 MPLS Failure Recovery and Convergence

To evaluate MPLS recovery behavior, the primary MPLS path between R1 and R3 was disabled.

The following command was applied on R1:

```
interface FastEthernet1/1
shutdown
```

After the failure:

- OSPF recalculated the routing table;
- LDP sessions were updated;
- MPLS forwarding switched to an alternative LSP.

Traceroute after the failure produced the following output:

```
1 192.168.5.2
2 192.168.5.6
3 192.168.5.14
4 192.168.5.26
5 192.168.5.30
```

The new path corresponded to:

$$R1 \to R2 \to R3 \to R5$$

This demonstrated successful MPLS rerouting and network resilience after link failure.

The failed interface was restored using:

```
interface FastEthernet1/1
no shutdown
```

After convergence, the original LSP was automatically restored.

3.9 Conclusion

This part of the laboratory successfully demonstrated the deployment and operation of MPLS forwarding in a Cisco-based network.

The experiments confirmed:

- successful LDP neighbor establishment;
- MPLS label distribution and forwarding;
- operation of Penultimate Hop Popping (PHP);
- MPLS traffic analysis using Wireshark;
- MPLS recovery after link failures.

The obtained results demonstrated the robustness and flexibility of MPLS networks under dynamic routing conditions.

Overall, the laboratory provided practical experience with MPLS forwarding, label switching, and traffic engineering concepts.

4 Part C — MPLS VPN Core Network

4.1 Introduction

The objective of Part C was to design and implement a redundant MPLS provider backbone capable of supporting MPLS VPN services.

Compared to the previous laboratory parts, the network topology was significantly expanded in order to simulate a realistic operator network with multiple Provider Edge (PE) and core Label Switch Routers (LSRs).

The experiments performed in this part focused on:

- implementation of a redundant MPLS core;
- OSPF routing in the provider backbone;
- MPLS forwarding and LDP operation;
- MPLS forwarding table analysis;
- MPLS path identification;
- MPLS failover and recovery;
- traffic rerouting after link failures.

4.2 Provider Network Topology

The MPLS provider network was composed of:

- 3 PE routers;
- 5 P routers;
- multiple redundant MPLS paths.

The topology was designed to guarantee high redundancy and fault tolerance.

The following routers were used:

- PE routers: R1-PE, R7-PE and R8-PE;
- P routers: R2, R3, R4, R5 and R6.

Multiple redundant paths were implemented between PE routers to guarantee MPLS connectivity even after internal link failures.

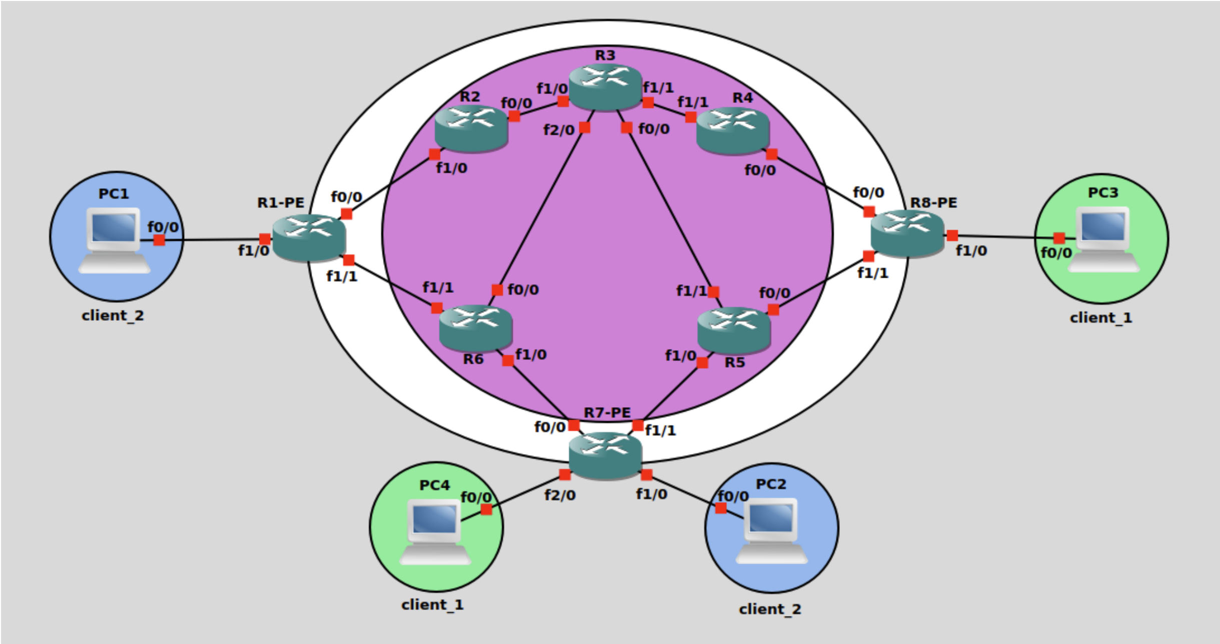


Figure 4: Redundant MPLS provider backbone topology

4.3 IPv4 Addressing Plan

All inter-router links were configured using dedicated /30 point-to-point subnets.

Link	Subnet
R1-R2	10.0.12.0/30
R1-R6	10.0.16.0/30
R2-R3	10.0.23.0/30
R3-R4	10.0.34.0/30
R3-R5	10.0.37.0/30
R3-R6	10.0.36.0/30
R4-R8	10.0.48.0/30
R5-R8	10.0.58.0/30
R5-R7	10.0.57.0/30
R6-R7	10.0.67.0/30

Table 2: IPv4 addressing plan used in the MPLS provider backbone

Loopback interfaces were also configured on all routers and used as router identifiers.

4.4 OSPF Configuration

OSPF Area 0 was configured throughout the provider backbone to distribute routing information between all MPLS routers.

Example configuration:

```
router ospf 1
 network 10.0.0.0 0.255.255.255 area 0
```

Loopback interfaces were also advertised through OSPF to guarantee end-to-end reachability between PE routers.

OSPF neighbor relationships were verified using:

```
show ip ospf neighbor
```

Example output obtained on R5:

```
Neighbor ID     State      Address
7.7.7.7         FULL       10.0.57.2
3.3.3.3         FULL       10.0.37.1
8.8.8.8         FULL       10.0.58.1
```

This confirmed successful OSPF adjacency establishment across the redundant backbone.

4.5 MPLS and LDP Configuration

Cisco Express Forwarding (CEF) was enabled on all routers before activating MPLS.

The following commands were applied globally:

```
ip cef
mpls ip
```

MPLS was then enabled on all core interfaces.

Example configuration on R3:

```
interface FastEthernet0/0
 mpls ip

interface FastEthernet1/0
 mpls ip

interface FastEthernet1/1
 mpls ip
```

LDP neighbor relationships were verified using:

```
show mpls ldp neighbor
```

The output confirmed successful label exchange between MPLS routers.

4.6 MPLS Forwarding Validation

MPLS forwarding operation was validated using:

```
show mpls forwarding-table
```

Example output:

```
Local         Outgoing     Prefix
Label         Label

20            Pop Label    10.0.58.0/30
22            27           8.8.8.8/32
```

The forwarding table confirmed:

- MPLS label distribution;
- label switching operation;
- usage of Penultimate Hop Popping (PHP).

4.7 MPLS Traceroute Analysis

Connectivity between PE routers was tested using ICMP and MPLS traceroute operations.
The following command was executed on R1-PE:

```
traceroute 8.8.8.8
```

Observed output before link failure:

```
1 10.0.12.2 [MPLS: Label 22 Exp 0]
2 10.0.36.1 [MPLS: Label 27 Exp 0]
3 10.0.34.2 [MPLS: Label 22 Exp 0]
4 10.0.48.2
```

This path corresponded to:

$$R1\text{-}PE \to R6 \to R3 \to R4 \to R8\text{-}PE$$

The MPLS labels visible in the traceroute confirmed successful MPLS forwarding across the provider backbone.

4.8 Wireshark MPLS Analysis

Wireshark was used to inspect MPLS packets traversing the network.
Captures performed inside the MPLS core showed MPLS headers similar to:

```
MultiProtocol Label Switching Header
Label: 22
```

This confirmed that MPLS labels were correctly attached and switched throughout the provider backbone.

Wireshark captures performed on the link between R4 and R8 showed MPLS encapsulated packets traversing the provider backbone.

The captured packets contained MPLS headers carrying label information before the IPv4 and ICMP headers.

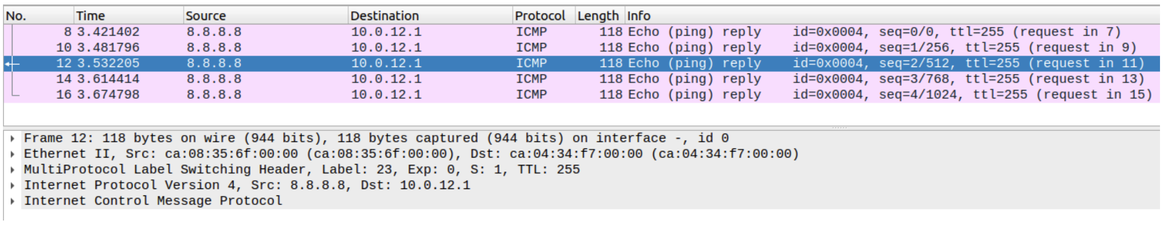


Figure 5: MPLS labeled packet captured inside the MPLS core

The MPLS forwarding table also showed the use of the `Pop Label` operation, demonstrating Penultimate Hop Popping (PHP).

4.9 MPLS Failure Recovery and Redundancy

One of the objectives of this part was to verify MPLS resiliency under link failure conditions. To simulate a network failure, the link between R3 and R4 was disabled.

The following command was executed on R3:

```
interface FastEthernet1/1
shutdown
```

After the failure:

- OSPF recalculated the routing tables;
- MPLS LSPs were recomputed;
- traffic was automatically redirected through an alternative path.

Connectivity remained operational during the failure:

```
!!!!!
Success rate is 100 percent (5/5)
```

Traceroute after convergence produced the following output:

```
1 10.0.12.2 [MPLS: Label 22 Exp 0]
2 10.0.36.1 [MPLS: Label 27 Exp 0]
3 10.0.37.2 [MPLS: Label 22 Exp 0]
4 10.0.58.1
```

The new MPLS path corresponded to:

$$R1\text{-}PE \rightarrow R6 \rightarrow R7\text{-}PE \rightarrow R5 \rightarrow R8\text{-}PE$$

This demonstrated:

- MPLS resiliency;
- automatic OSPF convergence;
- dynamic MPLS rerouting;

- successful failover operation.

After the test, the failed interface was restored using:

```
interface FastEthernet1/1
no shutdown
```

The original MPLS path was automatically re-established.

4.10 MPLS Layer 3 VPN Deployment

After validating the MPLS core operation, an MPLS Layer 3 VPN service was implemented between two remote customer sites.

The objective of this experiment was to verify the transport of customer traffic across the MPLS provider backbone using:

- Virtual Routing and Forwarding instances (VRFs);
- Multiprotocol BGP (MP-BGP);
- VPNv4 route exchange;
- MPLS label stacking.

Two customer sites belonging to `CLIENT_1` were connected:

- PC3 connected to R8-PE;
- PC4 connected to R7-PE.

4.10.1 VRF Configuration

A VRF named `CLIENT_1` was created on both PE routers.

The following parameters were configured:

- Route Distinguisher (RD);
- Route Targets (RT);
- VPN route import/export policies.

Example configuration:

```
ip vrf CLIENT_1
 rd 100:1
 route-target export 100:1
 route-target import 100:1
```

Customer-facing interfaces were then associated with the VRF.

Example on R7-PE:

```
interface FastEthernet2/0
 ip vrf forwarding CLIENT_1
 ip address 192.168.40.1 255.255.255.0
```

Example on R8-PE:

```
interface FastEthernet1/0
 ip vrf forwarding CLIENT_1
 ip address 192.168.30.1 255.255.255.0
```

4.10.2 MP-BGP VPNv4 Configuration

MP-BGP sessions were established between the PE routers using their loopback interfaces.

The VPNv4 address-family was activated to exchange VPN routes through the MPLS backbone.

Example configuration:

```
router bgp 65000

 neighbor 8.8.8.8 remote-as 65000
 neighbor 8.8.8.8 update-source Loopback0

 address-family vpnv4
  neighbor 8.8.8.8 activate
  neighbor 8.8.8.8 send-community extended
 exit-address-family

 address-family ipv4 vrf CLIENT_1
  redistribute connected
 exit-address-family
```

The MP-BGP VPNv4 session was successfully established between the PE routers.

```
R7-PE#sh ip bgp vpnv4 all summary
BGP router identifier 7.7.7.7, local AS number 65000
BGP table version is 4, main routing table version 4
2 network entries using 312 bytes of memory
2 path entries using 160 bytes of memory
2/2 BGP path/bestpath attribute entries using 304 bytes of memory
1 BGP extended community entries using 24 bytes of memory
0 BGP route-map cache entries using 0 bytes of memory
0 BGP filter-list cache entries using 0 bytes of memory
BGP using 800 total bytes of memory
BGP activity 2/0 prefixes, 2/0 paths, scan interval 60 secs

Neighbor        V           AS MsgRcvd MsgSent   TblVer  InQ OutQ Up/Down  State/PfxRcd
8.8.8.8         4        65000      22      22        4    0    0 00:15:46        1
```

Figure 6: MP-BGP VPNv4 session established between PE routers

4.10.3 VPNv4 Route Exchange

VPNv4 routes were successfully exchanged between PE routers.

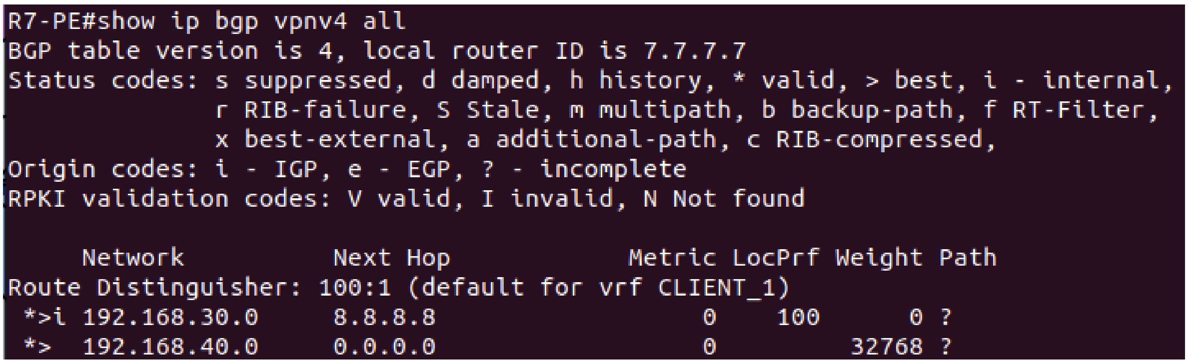
The following output obtained on R7-PE confirmed the reception of remote VPN routes from R8-PE:

```
Route Distinguisher: 100:1 (default for vrf CLIENT_1)

*>i 192.168.30.0       8.8.8.8
*>  192.168.40.0       0.0.0.0
```

This demonstrated that:

- remote customer routes were transported through MP-BGP;
- VRF route separation was operational;
- the MPLS backbone successfully transported VPNv4 information.



```
R7-PE#show ip bgp vpnv4 all
BGP table version is 4, local router ID is 7.7.7.7
Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
              r RIB-failure, S Stale, m multipath, b backup-path, f RT-Filter,
              x best-external, a additional-path, c RIB-compressed,
Origin codes: i - IGP, e - EGP, ? - incomplete
RPKI validation codes: V valid, I invalid, N Not found

     Network          Next Hop            Metric LocPrf Weight Path
Route Distinguisher: 100:1 (default for vrf CLIENT_1)
 *>i 192.168.30.0     8.8.8.8                  0    100      0 ?
 *>  192.168.40.0     0.0.0.0                  0         32768 ?
```

Figure 7: VPNv4 routes learned through MP-BGP

4.10.4 VRF Routing Table Verification

The VRF routing tables confirmed the successful installation of remote VPN routes learned through MP-BGP.

On R7-PE, the routing table associated with `CLIENT_1` contained:

- the locally connected customer network;
- the remote customer network learned through MP-BGP.

The routing table showed that traffic destined to the remote customer site was forwarded through the VPN next-hop associated with the remote PE router.

```
R7-PE#show ip route vrf CLIENT_1

Routing Table: CLIENT_1
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override

Gateway of last resort is not set

B     192.168.30.0/24 [200/0] via 8.8.8.8, 00:04:06
      192.168.40.0/24 is variably subnetted, 2 subnets, 2 masks
C        192.168.40.0/24 is directly connected, FastEthernet2/0
L        192.168.40.1/32 is directly connected, FastEthernet2/0
```

Figure 8: VRF routing table containing local and remote VPN routes

4.10.5 End-to-End VPN Connectivity

Connectivity between customer sites was validated using ICMP traffic between PC3 and PC4.

The following test was performed:

```
PC3# ping 192.168.40.2
```

The communication succeeded successfully across the MPLS provider backbone.

This confirmed:

- correct VRF operation;
- successful MP-BGP route distribution;
- MPLS VPN forwarding functionality;
- end-to-end customer connectivity.

4.10.6 VPN Traceroute Analysis

Traceroute operations performed between customer sites demonstrated that traffic successfully traversed the MPLS provider backbone.

The traceroute output showed MPLS labels being used while forwarding packets through intermediate provider routers.

Example observed labels:

- transport labels;
- VPN labels associated with the customer VRF.

This confirmed that customer traffic was transported transparently across the MPLS infrastructure while remaining isolated inside the VPN instance.

```
PC3#trace 192.168.40.2
Type escape sequence to abort.
Tracing the route to 192.168.40.2
VRF info: (vrf in name/id, vrf out name/id)
  1 192.168.30.1 16 msec 28 msec 32 msec
  2 10.0.58.2 [MPLS: Labels 30/17 Exp 0] 48 msec 40 msec 32 msec
  3 192.168.40.1 44 msec 40 msec 32 msec
  4 192.168.40.2 72 msec 60 msec 60 msec
```

Figure 9: Traceroute between VPN customer sites through the MPLS backbone

4.10.7 Wireshark MPLS VPN Analysis

Wireshark captures performed inside the MPLS core confirmed the successful transport of VPN traffic using MPLS encapsulation.

The captured packets showed MPLS label stacking, where:

- the outer label was used for MPLS transport through the provider backbone;
- the inner label identified the VPN associated with the customer VRF.

The captured packets also contained IPv4 and ICMP headers corresponding to customer communication between remote VPN sites.

Example packet structure observed:

```
Ethernet II
MPLS Label 19
MPLS Label 16
IPv4 Src: 192.168.40.2
IPv4 Dst: 192.168.30.2
ICMP Echo Reply
```

This confirmed the successful implementation of MPLS Layer 3 VPN services over the MPLS provider network.

No.	Time	Source	Destination	Protocol	Length	Info
17	13.975322	7.7.7.7	8.8.8.8	BGP	77	KEEPALIVE Message
29	26.236696	192.168.30.2	192.168.40.2	ICMP	118	Echo (ping) request id=0x0003, seq=0/0, ttl=254 (reply in 30)
30	26.248431	192.168.40.2	192.168.30.2	ICMP	122	Echo (ping) reply id=0x0003, seq=0/0, ttl=254 (request in 29)
31	26.297684	192.168.30.2	192.168.40.2	ICMP	118	Echo (ping) request id=0x0003, seq=1/256, ttl=254 (reply in 32)
32	26.346827	192.168.40.2	192.168.30.2	ICMP	122	Echo (ping) reply id=0x0003, seq=1/256, ttl=254 (request in 31)
33	26.401116	192.168.30.2	192.168.40.2	ICMP	118	Echo (ping) request id=0x0003, seq=2/512, ttl=254 (reply in 34)
34	26.430654	192.168.40.2	192.168.30.2	ICMP	122	Echo (ping) reply id=0x0003, seq=2/512, ttl=254 (request in 33)
35	26.461833	192.168.30.2	192.168.40.2	ICMP	118	Echo (ping) request id=0x0003, seq=3/768, ttl=254 (reply in 36)
36	26.480403	192.168.40.2	192.168.30.2	ICMP	122	Echo (ping) reply id=0x0003, seq=3/768, ttl=254 (request in 35)
37	26.504358	192.168.30.2	192.168.40.2	ICMP	118	Echo (ping) request id=0x0003, seq=4/1024, ttl=254 (reply in 38)
38	26.520764	192.168.40.2	192.168.30.2	ICMP	122	Echo (ping) reply id=0x0003, seq=4/1024, ttl=254 (request in 37)

```
Frame 34: 122 bytes on wire (976 bits), 122 bytes captured (976 bits) on interface -, id 0
Ethernet II, Src: ca:07:35:51:00:1d (ca:07:35:51:00:1d), Dst: ca:05:35:15:00:1c (ca:05:35:15:00:1c)
MultiProtocol Label Switching Header, Label: 19, Exp: 0, S: 0, TTL: 254
MultiProtocol Label Switching Header, Label: 16, Exp: 0, S: 1, TTL: 254
Internet Protocol Version 4, Src: 192.168.40.2, Dst: 192.168.30.2
Internet Control Message Protocol
```

Figure 10: Wireshark capture showing MPLS VPN label stacking and VPN traffic encapsulation

4.11 Multiple VPN Instances and Overlapping Addressing

After validating the operation of a single MPLS Layer 3 VPN instance, a second customer VPN was deployed over the same MPLS provider backbone.

The objective of this experiment was to demonstrate:

- simultaneous operation of multiple VPNs;
- VPN isolation using VRFs;
- MP-BGP support for multiple customer routing tables;
- support for overlapping customer IPv4 addressing.

A second VRF named `CLIENT_2` was created on the provider edge routers.

The following configuration was applied:

```
ip vrf CLIENT_2
 rd 200:1
 route-target export 200:1
 route-target import 200:1
```

The second customer was connected using:

- PC1 connected to R1-PE;
- PC2 connected to R7-PE.

4.11.1 Overlapping VPN Address Spaces

To demonstrate MPLS VPN isolation, the same IPv4 prefixes used previously in `CLIENT_1` were reused in `CLIENT_2`.

The following networks were configured:

- 192.168.30.0/24
- 192.168.40.0/24

Although both VPNs transported identical customer prefixes, MPLS VPN separation was successfully achieved using:

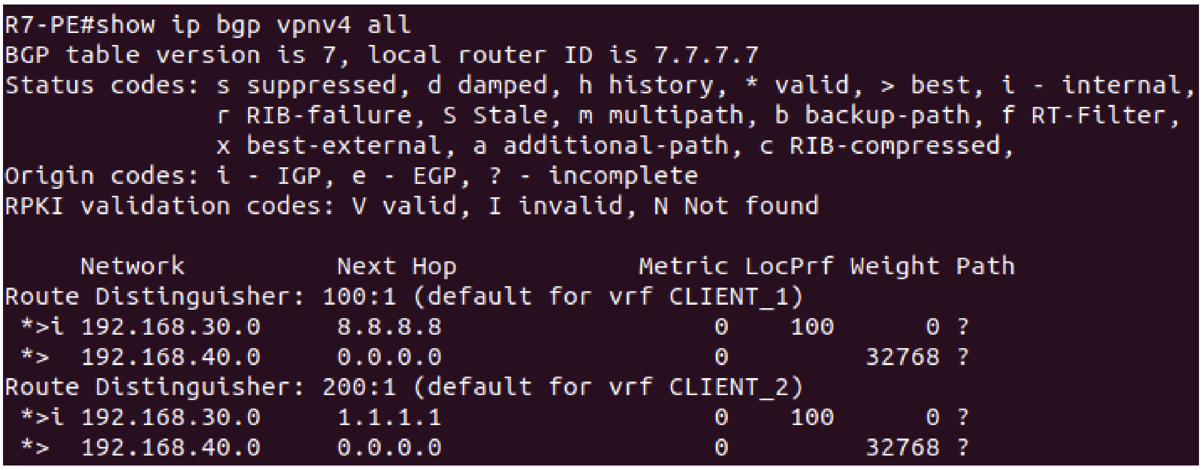
- distinct VRFs;
- independent routing tables;
- different Route Distinguishers (RDs);
- MP-BGP VPNv4 route separation.

The following output confirmed the coexistence of overlapping routes inside different VPN instances:

```
Route Distinguisher: 100:1
 *>i 192.168.30.0      8.8.8.8
 *>  192.168.40.0      0.0.0.0

Route Distinguisher: 200:1
 *>i 192.168.30.0      1.1.1.1
 *>  192.168.40.0      0.0.0.0
```

This demonstrated that identical IPv4 prefixes could coexist inside the MPLS backbone while remaining logically separated through VPNv4 routing.



```
R7-PE#show ip bgp vpnv4 all
BGP table version is 7, local router ID is 7.7.7.7
Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
              r RIB-failure, S Stale, m multipath, b backup-path, f RT-Filter,
              x best-external, a additional-path, c RIB-compressed,
Origin codes: i - IGP, e - EGP, ? - incomplete
RPKI validation codes: V valid, I invalid, N Not found

     Network          Next Hop            Metric LocPrf Weight Path
Route Distinguisher: 100:1 (default for vrf CLIENT_1)
 *>i 192.168.30.0     8.8.8.8                  0    100      0 ?
 *>  192.168.40.0     0.0.0.0                  0         32768 ?
Route Distinguisher: 200:1 (default for vrf CLIENT_2)
 *>i 192.168.30.0     1.1.1.1                  0    100      0 ?
 *>  192.168.40.0     0.0.0.0                  0         32768 ?
```

Figure 11: Multiple VPNv4 routing tables with overlapping customer prefixes

4.11.2 Multiple VRF Operation on a Shared PE Router

One of the most important aspects of this experiment was the simultaneous operation of multiple VRFs on the same Provider Edge router.

R7-PE simultaneously served:

- `CLIENT_1`;
- `CLIENT_2`.

This demonstrated how MPLS Layer 3 VPN services allow a single provider router to support multiple independent customers while maintaining traffic separation.

The VRF routing tables confirmed that customer routes were correctly separated and associated with their respective VPN instances.

```
R7-PE#show ip route vrf CLIENT_2

Routing Table: CLIENT_2
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override

Gateway of last resort is not set

B     192.168.30.0/24 [200/0] via 1.1.1.1, 00:06:05
      192.168.40.0/24 is variably subnetted, 2 subnets, 2 masks
C        192.168.40.0/24 is directly connected, FastEthernet1/0
L        192.168.40.1/32 is directly connected, FastEthernet1/0
```

Figure 12: VRF routing table associated with CLIENT_2

4.11.3 VPN Connectivity Validation

Connectivity tests confirmed successful MPLS VPN operation between the second customer sites.

The following test was successfully performed:

```
PC1# ping 192.168.40.2
!!!!!
```

This confirmed:

- successful MPLS VPN forwarding;
- MP-BGP route exchange;
- VRF-based traffic separation;
- simultaneous operation of multiple customer VPNs.

4.11.4 Observed Limitation During Overlapping Addressing Tests

During the overlapping addressing experiments, an additional behavior was observed due to the use of router-based hosts inside the GNS3 environment.

Because identical IPv4 addresses were reused in multiple customer segments connected through the same virtual Ethernet environment, some ICMP traffic was locally resolved through ARP without traversing the MPLS VPN infrastructure.

As a result, certain connectivity tests produced responses directly at Layer 2 instead of being transported through the MPLS backbone.

This behavior occurred because:

- routers were being used as simulated PCs;

- identical IPv4 addresses existed simultaneously in connected Ethernet domains;
- local ARP resolution occurred before MPLS forwarding.

Despite this limitation, the MPLS VPN control plane operated correctly, as demonstrated by:

- VPNv4 routing tables;
- distinct Route Distinguishers;
- successful MP-BGP route exchange;
- independent VRF routing tables.

Therefore, the experiment still successfully demonstrated MPLS Layer 3 VPN isolation and overlapping address support.

5 MPLS Traffic Engineering (MPLS-TE)

After validating the operation of the MPLS backbone, VPN services, and MP-BGP route exchange, MPLS Traffic Engineering (MPLS-TE) was implemented to demonstrate explicit traffic steering across the provider network.

The objective of this experiment was to:

- create MPLS-TE tunnels between Provider Edge routers;
- force traffic through explicit paths different from the default IGP shortest path;
- demonstrate RSVP signalling and TE tunnel establishment;
- validate unidirectional MPLS-TE forwarding paths.

5.1 MPLS-TE Configuration

MPLS Traffic Engineering was enabled on all provider routers participating in the MPLS core.

The following features were configured:

- RSVP signalling;
- MPLS-TE on core interfaces;
- TE router identifiers using loopback interfaces;
- explicit MPLS paths;
- MPLS-TE tunnel interfaces.

The OSPF process was extended with TE support:

```
router ospf 1
 mpls traffic-eng router-id Loopback0
 mpls traffic-eng area 0
```

Traffic Engineering was then activated on all MPLS backbone interfaces:

```
interface FastEthernet0/0
 ip rsvp bandwidth 100000
 mpls traffic-eng tunnels
```

5.2 Explicit Traffic Engineering Paths

Two unidirectional MPLS-TE tunnels were configured between the PE routers:

- R1-PE → R8-PE
- R8-PE → R1-PE

Each tunnel followed an explicitly configured path diverging from the default IGP SPF calculation.

The outbound tunnel from R1-PE to R8-PE was configured to follow:

$$R1 \rightarrow R2 \rightarrow R3 \rightarrow R4 \rightarrow R8$$

The return tunnel from R8-PE to R1-PE followed:

$$R8 \rightarrow R5 \rightarrow R7 \rightarrow R6 \rightarrow R1$$

Explicit MPLS-TE paths were defined using:

```
ip explicit-path name R1_TO_R8 enable
 next-address 2.2.2.2
 next-address 3.3.3.3
 next-address 4.4.4.4
 next-address 8.8.8.8
```

The tunnel interfaces were then configured using MPLS Traffic Engineering mode:

```
interface Tunnel0
 ip unnumbered Loopback0
 tunnel mode mpls traffic-eng
 tunnel destination 8.8.8.8
 tunnel mpls traffic-eng path-option 1 explicit name R1_TO_R8
 tunnel mpls traffic-eng autoroute announce
```

5.3 Tunnel Establishment Validation

The following command confirmed successful MPLS-TE tunnel establishment:

```
show mpls traffic-eng tunnels
```

The output showed:

- operational MPLS-TE tunnels;
- active explicit paths;
- RSVP signalling information;
- MPLS label allocation;
- successful LSP establishment.

```
Name: R1-PE_t0                              (Tunnel0) Destination: 8.8.8.8
  Status:
    Admin: up         Oper: up     Path: valid       Signalling: connected
    path option 1, type explicit R1_TO_R8 (Basis for Setup, path weight 4)

  Config Parameters:
    Bandwidth: 0        kbps (Global)  Priority: 7  7   Affinity: 0x0/0xFFFF
    Metric Type: TE (default)
    AutoRoute:  enabled   LockDown: disabled  Loadshare: 0        bw-based
    auto-bw: disabled
  Active Path Option Parameters:
    State: explicit path option 1 is active
    BandwidthOverride: disabled  LockDown: disabled  Verbatim: disabled


  InLabel  :  -
  OutLabel : FastEthernet0/0, 17
  RSVP Signalling Info:
       Src 1.1.1.1, Dst 8.8.8.8, Tun_Id 0, Tun_Instance 1
    RSVP Path Info:
      My Address: 10.0.12.1
      Explicit Route: 10.0.12.2 10.0.23.1 10.0.23.2 10.0.34.1
                      10.0.34.2 10.0.48.1 10.0.48.2 8.8.8.8
      Record   Route:   NONE
      Tspec: ave rate=0 kbits, burst=1000 bytes, peak rate=0 kbits
    RSVP Resv Info:
      Record   Route:   NONE
      Fspec: ave rate=0 kbits, burst=1000 bytes, peak rate=0 kbits
  History:
    Tunnel:
      Time since created: 2 minutes, 47 seconds
      Time since path change: 2 minutes, 46 seconds
      Number of LSP IDs (Tun_Instances) used: 1
    Current LSP:
      Uptime: 2 minutes, 46 seconds

LSP Tunnel R8-PE_t0 is signalled, connection is up
  InLabel  : FastEthernet1/1, implicit-null
  OutLabel :  -
  RSVP Signalling Info:
       Src 8.8.8.8, Dst 1.1.1.1, Tun_Id 0, Tun_Instance 1
    RSVP Path Info:
      My Address: 1.1.1.1
      Explicit Route:  NONE
      Record   Route:   NONE
      Tspec: ave rate=0 kbits, burst=1000 bytes, peak rate=0 kbits
    RSVP Resv Info:
      Record   Route:   NONE
      Fspec: ave rate=0 kbits, burst=1000 bytes, peak rate=0 kbits
```

Figure 13: Operational MPLS-TE tunnel with explicit routing path

The explicit path information confirmed that the TE tunnel was traversing the manually selected routers instead of relying solely on the default OSPF shortest-path calculation.

5.4 Traffic Steering Through MPLS-TE Tunnels

Traffic forwarding was then redirected through the MPLS-TE tunnel using the `autoroute announce` feature.

The routing table confirmed that traffic destined to R8-PE was now being forwarded through the TE tunnel interface:

```
Routing entry for 8.8.8.8/32
 Known via "ospf 1"
 via Tunnel0
```

```
R1-PE#sh ip route 8.8.8.8
Routing entry for 8.8.8.8/32
  Known via "ospf 1", distance 110, metric 5, type intra area
  Last update from 8.8.8.8 on Tunnel0, 00:04:24 ago
  Routing Descriptor Blocks:
  * 8.8.8.8, from 8.8.8.8, 00:04:24 ago, via Tunnel0
      Route metric is 5, traffic share count is 1
```

Figure 14: Routing table showing forwarding through the MPLS-TE tunnel

This demonstrated successful MPLS-TE traffic steering.

5.5 Traffic Engineering Path Verification

Traceroute tests were performed to validate the operational TE forwarding path.

The following traceroute confirmed that traffic from R1-PE to R8-PE was traversing the explicitly configured MPLS-TE path:

```
trace 8.8.8.8
```

The traceroute output confirmed traversal through:

$$R1 \rightarrow R2 \rightarrow R3 \rightarrow R4 \rightarrow R8$$

instead of using the default shortest OSPF route.

```
R1-PE#trace 8.8.8.8
Type escape sequence to abort.
Tracing the route to 8.8.8.8
VRF info: (vrf in name/id, vrf out name/id)
  1 10.0.12.2 [MPLS: Label 17 Exp 0] 52 msec 40 msec 80 msec
  2 10.0.23.2 [MPLS: Label 19 Exp 0] 92 msec 72 msec 68 msec
  3 10.0.34.2 [MPLS: Label 17 Exp 0] 76 msec 64 msec 76 msec
  4 10.0.48.2 76 msec 72 msec 68 msec
```

Figure 15: Traceroute demonstrating MPLS-TE explicit path forwarding

5.6 RSVP Signalling Validation

RSVP neighbor relationships were also validated.

The following command confirmed RSVP signalling adjacency establishment between MPLS core routers:

show ip rsvp neighbor

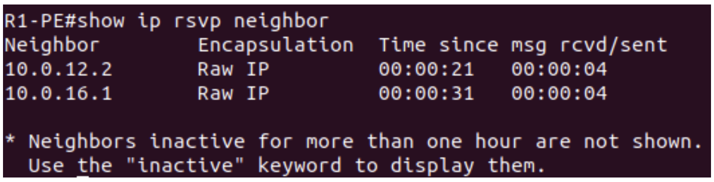


Figure 16: RSVP signaling neighbors inside the MPLS backbone

This demonstrated that RSVP signalling was operational and capable of supporting MPLS Traffic Engineering LSP establishment.

5.7 Wireshark MPLS-TE Analysis

Packet captures were collected using Wireshark in order to analyze MPLS-TE forwarding behavior inside the provider backbone.

Traffic captures performed between routers R2 and R3 confirmed that outbound traffic from R1-PE to R8-PE traversed the explicitly engineered TE path.

The captured packets showed:

- MPLS label encapsulation;
- ICMP payload transport;
- MPLS forwarding inside the TE tunnel.

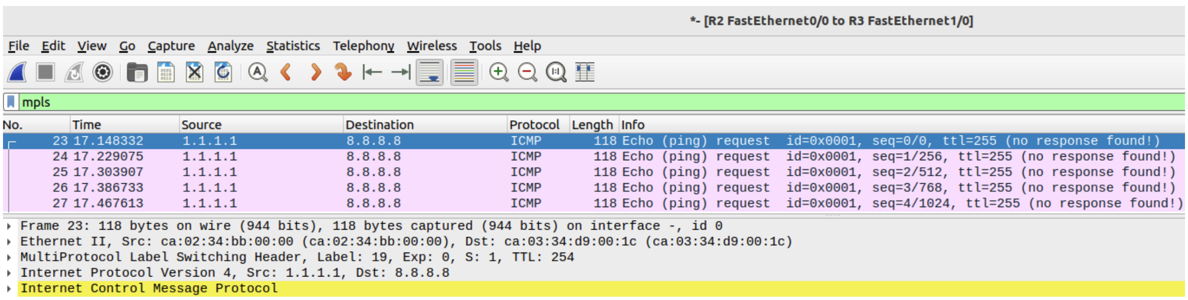


Figure 17: Outbound MPLS-TE traffic captured between R2 and R3

A second capture was performed between routers R6 and R1 to analyze the return traffic path.

This capture demonstrated that the return traffic followed a different TE tunnel path, validating the operation of unidirectional MPLS-TE tunnels.

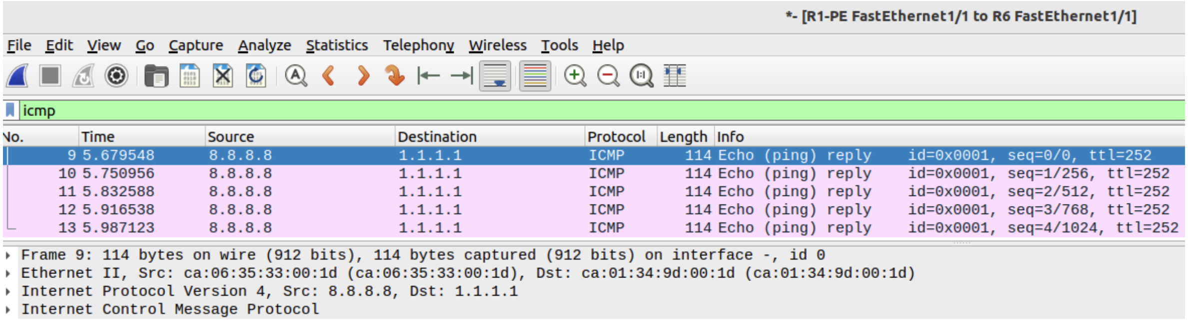


Figure 18: Inbound MPLS-TE traffic captured between R6 and R1

The packet captures confirmed:

- successful MPLS label switching;
- explicit traffic engineering operation;
- independent unidirectional TE tunnels;
- successful MPLS encapsulation throughout the provider backbone.

5.8 Constraint-Based Shortest Path First (CSPF)

An additional MPLS-TE tunnel was configured using dynamic path computation based on Constraint-Based Shortest Path First (CSPF).

Unlike the previous explicitly configured MPLS-TE tunnels, this tunnel dynamically computed its forwarding path using TE topology information distributed through OSPF-TE and RSVP signaling.

The tunnel was configured using a dynamic path option:

```
interface Tunnel1
 tunnel mode mpls traffic-eng
 tunnel destination 8.8.8.8
 tunnel mpls traffic-eng path-option 1 dynamic
```

The resulting tunnel output confirmed dynamic CSPF computation through the appearance of the `Shortest Unconstrained Path` information.

```
R1-PE#show mpls traffic-eng tunnels tunnel1

Name: R1-PE_t1                            (Tunnel1) Destination: 8.8.8.8
  Status:
    Admin: up         Oper: up     Path: valid       Signalling: connected
    path option 1, type dynamic (Basis for Setup, path weight 4)

  Config Parameters:
    Bandwidth: 0        kbps (Global)  Priority: 7  7   Affinity: 0x0/0xFFFF
    Metric Type: TE (default)
    AutoRoute:  enabled   LockDown: disabled  Loadshare: 0        bw-based
    auto-bw: disabled
  Active Path Option Parameters:
    State: dynamic path option 1 is active
    BandwidthOverride: disabled  LockDown: disabled  Verbatim: disabled


  InLabel  :  -
  OutLabel : FastEthernet0/0, 33
  RSVP Signalling Info:
       Src 1.1.1.1, Dst 8.8.8.8, Tun_Id 1, Tun_Instance 1
    RSVP Path Info:
      My Address: 10.0.12.1
      Explicit Route: 10.0.12.2 10.0.23.1 10.0.23.2 10.0.34.1
                      10.0.34.2 10.0.48.1 10.0.48.2 8.8.8.8
      Record   Route:   NONE
      Tspec: ave rate=0 kbits, burst=1000 bytes, peak rate=0 kbits
    RSVP Resv Info:
      Record   Route:   NONE
      Fspec: ave rate=0 kbits, burst=1000 bytes, peak rate=0 kbits
  Shortest Unconstrained Path Info:
    Path Weight: 4 (TE)
    Explicit Route: 10.0.12.1 10.0.12.2 10.0.23.1 10.0.23.2
                    10.0.34.1 10.0.34.2 10.0.48.1 10.0.48.2
                    8.8.8.8
  History:
    Tunnel:
      Time since created: 9 seconds
      Time since path change: 9 seconds
      Number of LSP IDs (Tun_Instances) used: 1
    Current LSP:
      Uptime: 9 seconds
```

Figure 19: Dynamic MPLS-TE tunnel using CSPF path computation

Traceroute validation confirmed that traffic was successfully transported through the dynamically computed MPLS-TE tunnel while preserving MPLS label switching.

```
R1-PE#   trace 8.8.8.8
Type escape sequence to abort.
Tracing the route to 8.8.8.8
VRF info: (vrf in name/id, vrf out name/id)
  1 10.0.12.2 [MPLS: Label 17 Exp 0] 72 msec 92 msec 60 msec
  2 10.0.23.2 [MPLS: Label 18 Exp 0] 80 msec 72 msec 72 msec
  3 10.0.34.2 [MPLS: Label 17 Exp 0] 72 msec 72 msec 68 msec
  4 10.0.48.2 92 msec 68 msec 76 msec
```

Figure 20: Traceroute through the dynamically computed CSPF MPLS-TE tunnel

This experiment demonstrated the operation of CSPF-based MPLS Traffic Engineering using dynamic TE path computation.

5.9 Conclusion

The MPLS Traffic Engineering implementation successfully demonstrated explicit traffic steering over the MPLS provider backbone.

The experiment validated:

- RSVP-based MPLS-TE signaling;
- explicit path selection;
- MPLS tunnel establishment;
- traffic steering through TE tunnels;
- MPLS forwarding using engineered paths;
- bidirectional traffic transport using independent unidirectional TE tunnels.

Together with the previously implemented MPLS VPN infrastructure, this final stage completed a fully operational MPLS provider network supporting:

- MPLS forwarding;
- MPLS VPN services;
- MP-BGP VPNv4 route exchange;
- VRF-based customer isolation;
- MPLS failover and reconvergence;
- MPLS Traffic Engineering.

6 Conclusion

This laboratory assignment successfully demonstrated the deployment and analysis of an OSPF-based IP network using Cisco routers in GNS3.

The implementation of VLSM enabled efficient IPv4 address allocation across all network segments. OSPF dynamically exchanged routing information and automatically selected optimal paths according to interface metrics.

Several experiments were performed to evaluate OSPF behavior under different network conditions, including:

- routing table analysis;
- OSPF cost manipulation;
- connectivity verification;
- link failure simulation;
- routing convergence analysis.

The obtained results demonstrated:

- successful end-to-end connectivity;
- dynamic path recalculation;
- automatic failover capabilities;
- fast OSPF convergence.

Overall, this assignment provided practical experience with dynamic routing protocols and fundamental traffic engineering concepts.